Mariah C. Boudreau

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Research Interests

Mathematical modeling for biological applications, stochastic modeling, probability generating functions, and public health applications.

Education

Ph.D. in Mathematical Sciences 2019 – 2024
University of Vermont, Burlington, VT
- Dissertation title: Probabilistic modeling of disease: Addressing uncertainties in within-host and population-level dynamics
- Advisors: Dr. Laurent Hébert-Dufresne and Dr. Chris Danforth

Bachelor of Science in Mathematics 2015 – 2019
Saint Michael's College, Colchester, VT
- Minors in computer science and statistics

Awards and Fellowships

Quantitative and Evolutionary STEM Training (QuEST) Program Trainee
August 2019 – August 2024
NSF Research Traineeship, University of Vermont, Burlington, VT

Research Experience

Postdoctoral Researcher September 2024 – Present
Network Science Institute and Roux Institute – Northeastern University, Portland, ME
Developing stochastic models to optimize disease surveillance and pathogen detection, in collaboration with the CDC and Gates Foundation. Analyzing multi-source surveillance data, including travel networks and voluntary testing surveys. Building reproducible, documented modeling tools that can be used by researchers in resource-limited settings.

Graduate Research Assistant June 2023 – August 2024
University of Vermont, Burlington, VT
Conducting data analysis on blood work data for the Lived Experience Measured Using Rings study.

Contractor May 2022 – July 2022
Institute for Disease Modeling at the Gates Foundation, Seattle, WA
Assisted in the parameterization and development of an open-source human papillomavirus population model. Began developing a within-host model for human papillomavirus induced lesions in a cervical skin layer.

Graduate Research Assistant August 2020 – May 2022
University of Vermont, Burlington, VT
Developed a project about the sensitivity of final outbreak predictions for probability generating functions (PGFs) when applied to epidemics. Another project expands on time-dependent PGFs to understand early epidemic uncertainty and incorporate intervention strategies.

Dartmouth Summer Research Student June 2018 – August 2018
Dartmouth College, Hanover, NH
Presented research on a simplified version of a multi-measurement vector problem.

BU SIBS Participant June 2017 – July 2017
Boston University, Boston, MA
Presented research analyzing data from the National Health Interview Survey of the CDC, particularly assessing the relationship between amounts of milk consumed during one's life was compared to being diagnosed with osteoporosis.

Other experience
Technical Services Intern, Mylan Technologies, Saint Albans, VT June 2019 – August 2019

Publications and other writings

M.C. Boudreau & G. St-Onge October 2026
Probability generating functions for epidemiological time series
arXiv

M.C. Boudreau, W.H.W Thompson, C.M. Danforth, J.G. Young, & L. Hébert-Dufresne February 2026
Sensitivity analysis of epidemic forecasting and spreading on networks with probability generating functions
J. R. Soc. Interface 23(235), 20250686

M.C. Boudreau, J.A. Cohen, & L. Hébert-Dufresne August 2024
Within-host infection dynamics with master equations and the method of moments: A case study of human papillomavirus in the epithelium
arXiv, under peer review at J. of Math. Biol.

R.M. Stuart, J.A. Cohen, C.C. Kerr, P. Mathur, R.G. Abeysuriya, M. Zimmerman, D.W. Rao, **M.C. Boudreau**, S. Lee, L. Yang, & D.J. Klein July 2024
HPVsim: An agent-based model of HPV transmission and cervical disease
PLOS Comput. Biol. 20(7) e1012181

M.C. Boudreau, A.J. Allen, N.J. Roberts, A. Allard, & L. Hébert-Dufresne September 2023
Temporal and probabilistic comparisons of epidemic interventions
Bull. of Math. Biol. 85, 118

A.J. Allen, **M.C. Boudreau**, N.J. Roberts, A. Allard, & L. Hébert-Dufresne February 2022
Predicting the diversity of early epidemic spread on networks
Phys. Rev. Research 4, 013123

Presentations

Generating function framework for spatiotemporal outbreak analytics June 2026
NetSci 2026 Conference, Boston, MA

Modeling and optimization of multimodal travel-based surveillance at airports December 2025
Epidemics10 Conference, San Diego, CA

Modeling disease spread with probability generating functions February 2025
Women in Network Science (WiNS) seminar, Online

Temporal and probabilistic comparisons of epidemic interventions June 2024
NetSci 2024 Conference, Québec City, QC, CA

Sensitivity analysis of stochastic polynomials, and its application to epidemic forecasting and random graphs August 2023
Contagions on Complex Social Systems Workshop, Burlington, VT

Temporal and probabilistic forecasts of epidemic interventions May 2023
SIAM Dynamical Systems 2023 Conference, Portland, OR

Temporal and probabilistic forecasts of epidemic interventions April 2023
University of Vermont Student Research Conference, Burlington, VT

Modeling epidemic interventions with probability generating functions April 2021
University of Vermont Virtual Student Research Conference, Online

Ski resort network analysis January 2021
Complex Networks Winter Workshop 2020, Online

Probabilistic epidemic forecasting using probability generating functions, and its robustness to data quality, error, biases, and noise September 2020
University of Vermont Computer Science Research Day, Online

Probabilistic epidemic forecasting using probability generating functions, and its robustness to data quality, error, biases, and noise September 2020
NetSci 2020 Conference, Online

Other conferences attended
Complex Networks Winter Workshop December 2023
Grace Hopper Celebration for Women in Tech September 2023
Women in Network Science Collabathon May 2023
NetSci 2019 June 2019

Teaching

Graduate Teaching Assistant August 2019 – May 2020 & August 2022 – May 2023
Mathematics Department, University of Vermont, Burlington, VT
Taught MATH 017: Applications of Finite Math in 2019–2020. Taught MATH 019: Fundamentals of Calculus I in 2022–2023.

QuEST Coding Workshop Teaching Assistant August 2021
University of Vermont, Burlington, VT
Coordinated and taught first year Ph.D. students the fundamentals of coding in R and calculus basics.

Service

Program Committee
NetSci 2026, Boston, MA June 2026

Peer Review

SIAM Journal on Life Sciences (SIALS)	September 2026
SIAM Journal on Applied Dynamical Systems	February 2026
Journal of Physics: Complexity	October 2025
Journal of Theoretical Biology	June 2025
Applied Network Science	April 2025
Journal of Theoretical Biology	March 2025

Joint Peer Review

Physical Review Research — July 2023

Volunteer Ski Patrol — March 2021 – Present

Assist with first-aid and mountain/event operations as an Outdoor Emergency Care technician at Bolton Valley Ski Resort and the New Hampshire Backcountry Patrol.

Outreach

Math 4 All Podcast — April 2023

Guest appearance discussing how we can understand the world as networks.

Network analysis and its application to biology — April 2023

University of Vermont MATHCounts, Burlington, VT

Network analysis and its application to biology — January 2021

University of Vermont, 4-H Teen Science Cafe, Online

Skills

Programming: Python (NumPy, pandas, Matplotlib, GeoPandas, scikit-learn), R, SQL, MATLAB, LaTeX, Java, C++

Computational: Stochastic simulation, high-performance computing, Git/GitHub, Jupyter Notebooks, functional and object-oriented programming, reproducible research workflows

Mathematical and Statistical: Stochastic and probabilistic modeling, branching processes, probability generating functions, compartmental and agent-based models, network analysis, master equations, Bayesian modeling, parameter estimation, uncertainty quantification, sensitivity analysis, statistical inference, machine learning

Languages: Conversational French

Other interests

Mountain adventures (hiking, skiing, and climbing), and reading.